

0.9% Sodium Chloride Injection

in VIAFLEX Plastic Container

Description

0.9% Sodium Chloride Injection is a sterile, nonpyrogenic solution for fluid and electrolyte replenishment in single dose containers for intravenous administration. It contains no antimicrobial agents. Composition, osmolarity, pH and ionic concentration are shown in Table 1.

The VIAFLEX plastic container is fabricated from a specially formulated polyvinyl chloride (PL 146® Plastic). The amount of water that can permeate from inside the container into the overwrap is insufficient to affect the solution significantly. Solutions in contact with the plastic container can leach out certain of its chemical components in very small amounts within the expiration period, e.g., di-2-ethylhexyl phthalate (DEHP), up to 5 parts per million. However, the safety of the plastic has been confirmed in tests in animals according to USP biological tests for plastic containers as well as by tissue culture toxicity studies.

Clinical Pharmacology

0.9% Sodium Chloride Injection has value as a source of water and electrolytes. It is capable of inducing diuresis depending on the clinical condition of the patient.

Indications and Usage

0.9% Sodium Chloride Injection is indicated as a source of water and electrolytes.

0.9% Sodium Chloride Injection is also indicated for use as a priming solution in hemodialysis procedures.

Contraindications

None known.

Warnings

0.9% Sodium Chloride Injection should be used with great care, if at all, in patients with congestive heart failure, severe renal insufficiency, and in clinical states in which there exists edema with sodium retention.

In patients with diminished renal function, administration of Sodium Chloride Injection may result in sodium retention.

Hypersensitivity reactions

- Hypersensitivity/infusion reactions, including hypotension, pyrexia, tremor, chills, urticaria, rash, and pruritus, have been reported with Sodium Chloride Injection.
- Stop the infusion immediately if signs or symptoms of hypersensitivity/infusion reactions develop. Appropriate therapeutic countermeasures must be instituted as clinically indicated.

Risk of Fluid and/or Solute Overload and Electrolyte Disturbances

Depending on the volume and rate of infusion, intravenous administration of Sodium Chloride Injection can cause:

- fluid and/or solute overload resulting in overhydration/hypervolemia and, for example, congested states, including central and peripheral edema.
- clinically relevant electrolyte disturbances and acid-base imbalance.

In general, the risk of dilutional states is inversely proportional to the electrolyte concentrations of 0.9% Sodium Chloride Injection and additions. The risk of solute overload causing congested states is directly proportional to the electrolyte concentrations of 0.9% Sodium Chloride Injection and additions.

Clinical evaluation and periodic laboratory determinations are necessary to monitor changes in fluid balance, electrolyte concentrations, and acid base balance during prolonged parenteral therapy or whenever the condition of the patient warrants such evaluation.

Use in patients at risk for sodium retention, fluid overload and edema

Sodium Chloride Injection should be used with particular caution, if at all, in patients with or at risk for:

- Hyponatremia
- Hyperchloremia
- Metabolic acidosis
- Hypervolemia
- Medications that may increase the risk of sodium and fluid retention, such as corticosteroids.
- Conditions that may cause sodium retention, fluid overload and edema (central and peripheral), such as patients with
 - primary hyperaldosteronism,
 - secondary hyperaldosteronism, associated with for example: hypertension, congestive heart failure, liver disease (including cirrhosis), renal disease (including renal artery stenosis, nephrosclerosis) or pre-eclampsia.

Risk of Hyponatremia

Monitoring of serum sodium is important for all fluids. 0.9% Sodium Chloride Injection has an osmolarity of 308 mOsmol/L (calc).

High volume infusion must be used under specific monitoring in patients with cardiac or pulmonary failure, and in patients with non-osmotic vasopressin release (including SIADH), due to the risk of hospital-acquired hyponatremia.

Acute hyponatremia can lead to acute hyponatremic encephalopathy (brain edema) characterized by headache, nausea, seizures, lethargy and vomiting. Patients with brain edema are at particular risk of severe, irreversible and life-threatening brain injury.

Use in Patients with Severe Renal Impairment

Sodium Chloride Injection should be administered with particular caution, if at all, to patients with severe renal impairment. In such patients, administration of Sodium Chloride Injection may result in sodium retention.

Precautions

Risk of Air Embolism

- Do not connect flexible plastic containers in series in order to avoid air embolism due to possible residual air contained in the primary container.
- Pressurizing intravenous solutions contained in flexible plastic containers to increase flow rates can result in air embolism if the residual air in the container is not fully evacuated prior to administration.
- Use of a vented intravenous administration set with the vent in the open position could result in air embolism. Vented intravenous administration sets with the vent in the open position should not be used with flexible plastic containers.

Interactions with Other Medicinal Products and Other Forms of Interaction

Caution is advised in patients treated with lithium. Renal sodium and lithium clearance may be increased during administration of Sodium Chloride Injection. Administration of Sodium Chloride Injection, may result in decreased lithium levels.

Regarding medications that increase risk of sodium and fluid retentions see section on Warnings. Caution is advised when administering 0.9% Sodium Chloride Injection to patients treated with drugs leading to an increased vasopressin effect. The below listed drugs increase the vasopressin effect, leading to reduced renal electrolyte free water excretion and may increase the risk of hyponatremia following treatment with i.v. fluids. (See sections on Warnings and Adverse Reactions).

Drugs stimulating vasopressin release such as chlorpropamide, clofibrate, carbamazepine, vincristine, selective serotonin reuptake inhibitors (SSRIs), 3,4-methylenedioxy-N-methamphetamine, ifosfamide, antipsychotics, opioids.

Drugs potentiating vasopressin action such as chlorpropamide, non-steroidal anti-inflammatories (NSAIDs), cyclophosphamide.

Vasopressin analogues such as desmopressin, oxytocin, vasopressin, terlipressin.

Caution is advised when administering 0.9% Sodium Chloride Injection to patients treated with drugs that may increase the risk of hyponatremia, such as diuretics and antiepileptics (e.g., oxcarbazepine).

Pregnancy, Lactation, and Fertility

There are no adequate data from the use of Sodium Chloride Injection in pregnant or lactating women. Physicians should carefully consider the potential risks and benefits for each specific patient before administering Sodium Chloride Injection.

Use in Pediatric Patients

Plasma electrolyte concentrations should be closely monitored in the pediatric population because of their impaired ability to regulate fluids and electrolytes.

Use in Geriatric Patients

When selecting the type of infusion solution and the volume/rate of infusion for a geriatric patient, consider that geriatric patients are generally more likely to have cardiac, renal, hepatic, and other diseases or concomitant drug therapy.

Adverse Reactions

Reactions which may occur because of the solution or technique of administration include febrile response, venous thrombosis or phlebitis extending from the site of injection, extravasation, and hypervolemia.

Post-marketing Adverse Reactions

- IMMUNE SYSTEM DISORDERS: Hypersensitivity/infusion reactions, including Hypotension, Pyrexia, Tremor, Chills, Urticaria, Rash, Pruritus
- GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS: Infusion site reactions, such as Infusion site erythema, Injection site streaking, Burning sensation, Infusion site urticaria.

Other Adverse Reactions / Class Reactions

The following adverse reactions have been reported with:

- Hyponatremia
- Hyperchloremic metabolic acidosis
- Hyponatremia, which may be symptomatic
- Hyponatremic encephalopathy

If an adverse reaction does occur, discontinue the infusion, evaluate the patient, institute appropriate therapeutic countermeasures and save the remainder of fluid for examination if deemed necessary.

Overdose

An excessive volume of 0.9% Sodium Chloride Injection may lead to hypernatremia (which can lead to CNS manifestations, including seizures, coma, cerebral edema and death) and sodium overload (which can lead to central and/or peripheral edema).

When assessing an overdose, any additives in the solution must also be considered. The effects of an overdose may require immediate medical attention and treatment.

Dosage and Administration

As directed by a physician. Dosage rate, and duration of administration are to be individualized and depend upon the indication for use, the patient's age, weight, clinical condition and concomitant treatment, and on the patient clinical and laboratory response to treatment.

When other electrolytes or medicines are added to this solution, the dosage and the infusion rate will also be dictated by the dose regimen of the additions.

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration whenever solution and container permit. Do not administer unless solution is clear and seal is intact.

All injection in VIAFLEX plastic containers are intended for intravenous administration using sterile equipment.

Additives may be incompatible. Complete information is not available. Those additives known to be incompatible should not be used. Consult with pharmacist, if available. If, in the informed judgment of the physician, it is deemed advisable to introduce additives, use aseptic technique. Before adding a substance or medication, verify that it is soluble and/or stable in water. The instructions for use of the medication to be added and other relevant literature must be consulted. After addition, check for a possible color change and/or the appearance of precipitates, insoluble complexes or crystals.

Mix thoroughly when additives have been introduced. Do not store solutions containing additives. For single use only. Discard any unused portion.

Availability

Table 1 shows the available sizes of each injection in VIAFLEX plastic containers.

Directions for use of VIAFLEX plastic container

Warning: Do not use plastic containers in series connections. Such use could result in air embolism due to residual air being drawn from the primary container before administration of the fluid from the secondary container is completed. See Precautions.

To Open

Tear overwrap down side at slit and remove solution container. Some opacity of the plastic due to moisture absorption during the sterilization process may be observed. This is normal and does not affect the solution quality or safety. The opacity will diminish gradually. Check for minute leaks by squeezing inner bag firmly. If leaks are found, discard solution as sterility may be impaired. If supplemental medication is desired, follow “to add medication” directions below

Preparation for Administration

- 1. Suspend container from eyelet support.
- 2. Remove plastic protector from outlet port at bottom of container.
- 3. Attach administration set. Refer to complete directions accompanying set.

To Add Medication

Warning: Additives known or determined to be incompatible should not be used. See Dosage and Administration.

To add medication before solution administration

- 1. Prepare medication site.
- 2. Using syringe with 19 to 22 gauge needle, puncture resealable medication port and inject.
- 3. Mix solution and medication thoroughly. For high density medication such as potassium chloride, squeeze ports while ports are upright and mix thoroughly.

Storage

Exposure of pharmaceutical products to heat should be minimized. Avoid excessive heat. It is recommended the product be stored at temperature not exceeding 30° C: brief exposure up to 40° C does not adversely affect the product.

To add medication during solution administration

- 1. Close clamp on the set.
- 2. Prepare medication site.
- 3. Using syringe with 19 to 22 gauge needle, puncture resealable medication port and inject.
- 4. Remove container from IV pole and/or turn to an upright position.
- 5. Evacuate both ports by squeezing them while container is in the upright position.
- 6. Mix solution and medication thoroughly.
- 7. Return container to in-use position and continue administration.

Caution :

Foods, Drugs, Devices, and Cosmetics Act prohibits dispensing without prescription.

Table 1	Sodium Chloride, (NaCl) (g/L)	Osmolarity (mOsmol/L) (calc)	Ionic Concentration (mEq/L)		Availability
			Sodium	Chloride	Size (mL)
0.9% Sodium Chloride Injection	9.0	308	154	154	50 100 250 500 1000